

E0.4 · SNR_SCALING · HBN THEPRESENT · R5 VAL / R6 TEST

The subject axis never saturated — the depth-12 encoder did.

Identical sweep, identical recipe, one change: depth 12 → depth 22 with a REVE warm start. The larger encoder lifts every readout, and keeps the scaling exponent alive past 700 subjects, where depth 12 goes flat. Each arm is measured against its own shape-matched random-init null.

depth 12 · 37.9M params · patch 400

depth 22 · 69.2M params · patch 200

FIGURE 1

Stimulus probe Δr^2 across the subject axis

Log-log in S. Bars are ± 1 sd across cohort draws where $n \geq 2$. The hollow point at S=1863 is a cell that never left chance loss; the filled point is its healthy re-run.

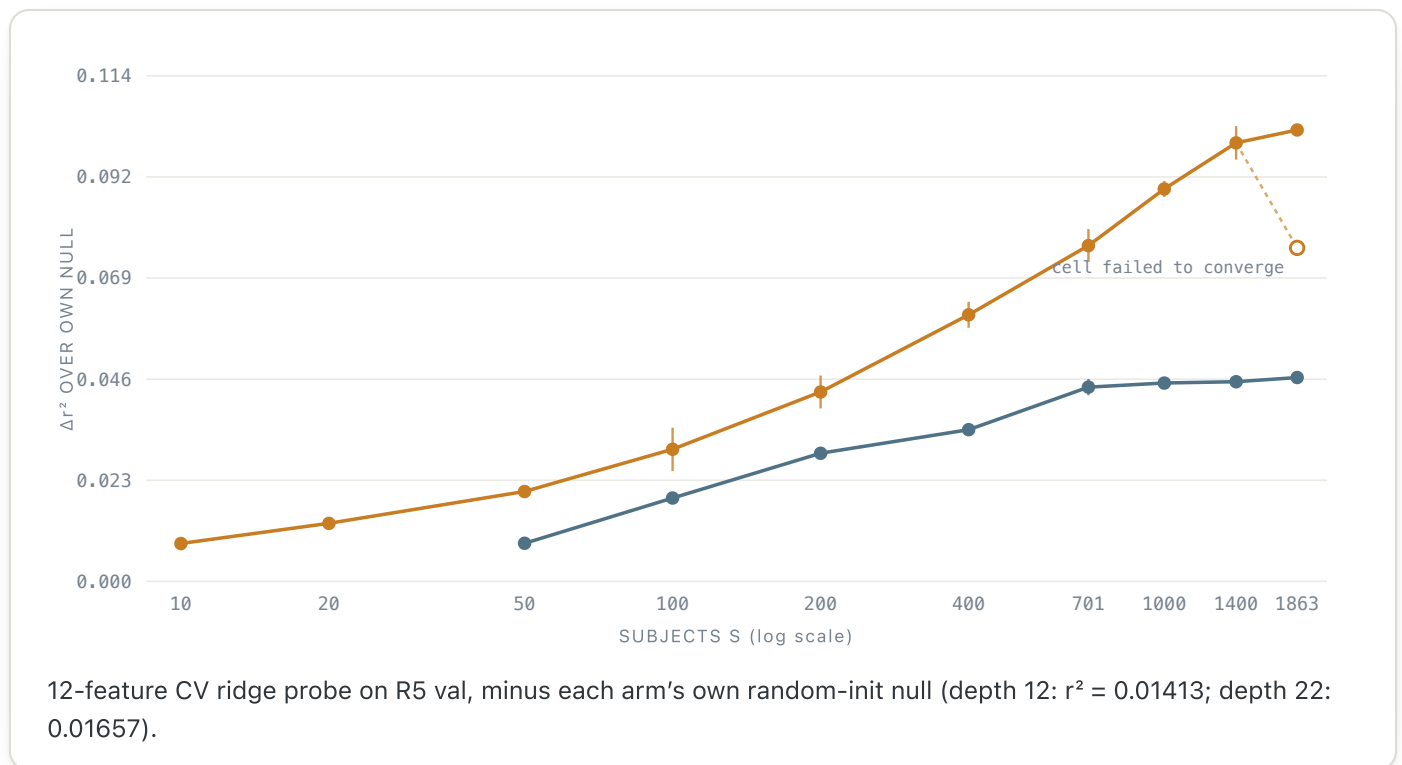
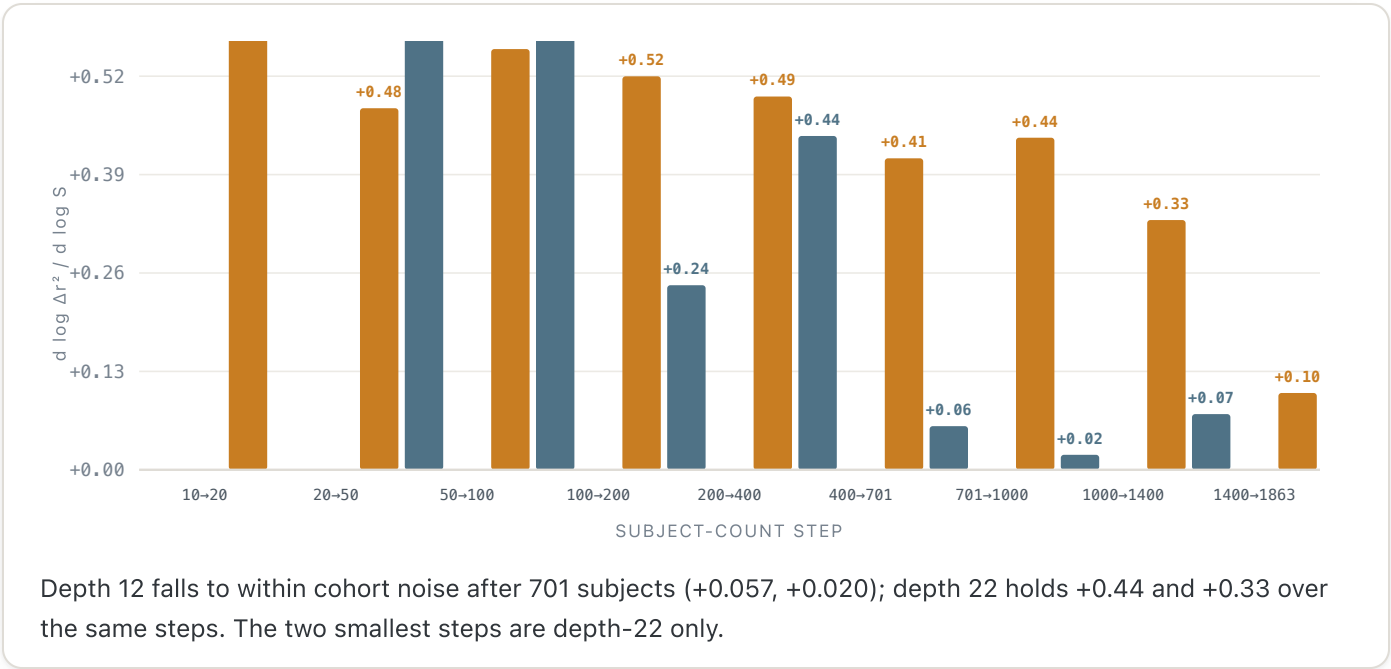


FIGURE 2

Local scaling exponent, step by step

$d \log \Delta r^2 / d \log S$ between adjacent scales — the quantity this project treats as the contribution. Both arms scale together up to ~700 subjects; only depth 22 keeps scaling past it.



UNDERLYING NUMBERS

Every readout measured on both arms

Same cells, same selected epochs. CC_norm is val *r* over the R5 single-trial ceiling of 0.313; retrieval is e→v top-1 at the time level on test, where both nulls sit at chance.

S	ΔR² (VAL)		TEST R		CC_NORM (VAL)		RETRIEVAL TOP-1	
	D12	D22	D12	D22	D12	D22	D12	D22
10	—	0.0086	—	0.1309	—	0.532	—	0.0147
20	—	0.0131	—	0.1391	—	0.582	—	0.0150
50	0.0086	0.0203	—	0.1509	—	0.635	—	0.0181
100	0.0189	0.0299	—	0.1742	—	0.713	—	0.0234
200	0.0290	0.0429	—	0.1980	—	0.801	—	0.0317
400	0.0343	0.0603	0.1729	0.2286	0.737	0.915	0.0333	0.0463
701	0.0440	0.0760	0.1865	0.2518	0.801	0.999	0.0351	0.0490
1000	0.0449	0.0888	0.1886	0.2775	0.813	1.066	0.0333	0.0581
1400	0.0452	0.0993	0.1918	0.2927	0.827	1.120	0.0362	0.0651
1863	0.0461	0.0755	0.1901	0.2579	0.822	0.992	0.0384	0.0523

PROVENANCE & GAPS

- **Nulls are not shared.** The depth-22 random encoder scores $r^2 = 0.01657$ against depth-12's 0.01413. Reusing the depth-12 null would have inflated every depth-22 Δr^2 by +0.0025 — larger than several real step sizes on this axis.
- **S < 400 is depth-12 only so far.** The matching depth-22 cells 15 queued are training; the figures extend left once they land. Neither arm has S=10 or S=20.
- **Three depth-22 cells landed in a bad optimisation basin** and were re-run at a different init seed — two stuck at exactly chance (InfoNCE loss $\ln 64 = 4.159$), one at S=200 with twice its siblings' loss and a quarter of their signal. Including the dead S=400 cell moved the 400→701 exponent from +0.411 to +0.982 — the difference between "slope unchanged" and "slope doubled".
- **S=1863 is n=1 in both arms by construction** — it is the whole subject pool, so no second cohort draw exists.
- **Depth-12's test r and retrieval columns are n=1** (only the first draw was ever read out), against n=3 for depth 22.
- **Depth 22 crosses CC_norm = 1 at S=1000**, i.e. it passes the measured single-trial val ceiling — the same signature that caused the test ceiling to be withdrawn as an underestimate.